

An Exploration of Nonclassical Carbocations

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I will present a journey that began with reading a paper describing a total synthesis of a natural product. In that paper, a reaction that I had not encountered before was mentioned – the Prins reaction. That reaction looked like a good candidate for Hammett plot experiments and I searched the literature for information on its mechanism and, hopefully, description of experiments and data. In several papers, a nonclassical carbocation intermediate was proposed. If you search the literature for the subject of nonclassical carbocations, you will quickly encounter stories of a dark time in physical organic chemistry when the giants went to war. This exploration will present the arguments and analyze the data that supported the victorious side of this tempest in our teacup.¹

My Path to the War

Synthetic organic chemistry is one place where the money is. It is wise to read papers in this field no matter what you are doing in chemistry. You will constantly find yourself exercising your skills in mechanism as you interpret synthetic schemes and marvel at clever solutions. My first step to one of the great battlegrounds of the 20th century started with this good habit of reading widely.

A Synthesis Plan

I was reading a recent paper from the group of Hanfeng Ding at Zhejiang Sci-Tech University in China.² In this contribution, the authors proposed a synthesis plan for melicolone A,³ a natural product isolated from a popular salad green and medicinal herb. In this scheme, they proposed using a Prins-type cyclization reaction. One should seek to understand *every* reaction in a scheme to expand one's toolbox of synthetic possibilities. I looked it up.

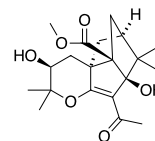
The Prins Reaction

The Prins reaction uses an electron-deficient carbon to perform an addition to an alkene.⁴ The original report from 1919 demonstrated the acid-catalyzed addition of formaldehyde to styrene. The moment I saw this, I sensed that a Hammett plot would have been done at some point to investigate this mechanism. I also see opportunities for experiments in kinetic isotope effects and perhaps there is data reported for the reaction in different acidity conditions where I could interpret a pH-rate profile using acidity functions.

I quickly found recent work from Benjamin List's group at the Max Planck Institute in Germany using chiral catalysts and paraformaldehyde to create chiral diols as building block for valuable pharmaceuticals.⁵

This document was produced using the L^AT_EX typesetting language with the Tufte-handout document class. Chemical diagrams were created in ChemDoodle. Plots were typeset using Matplotlib and Python tools. Diagrams were created and edited using Affinity Designer.

¹ We care about this because this tempest was in our teacup. The battle consumed the headspace of some of the greatest contributors to the field of physical organic chemistry for decades. In the end, the world did not change, and that was the problem. Was this argument worth the time?

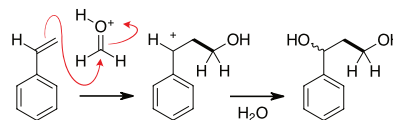


Melicolone A

² "A Biomimetic Approach for Construction of an Advanced Intermediate en Route to Melicolones A and B." T. Wang, Z. Jia, C. Zhu, L. Zhang, H. Ding, *J. Org. Chem.*, 2025, 90, 7531–7539. <https://doi.org/10.1021/acs.joc.5c00734>.

³ see Scheme 1 in reference 2

⁴ "Recent Advances in the Prins Reaction." E. Reyes, L. Prieto, U. Uribe, L. Carrillo, J.L. Vicario, *ACS Omega*, 2022, 7, 31621–31627. <https://doi.org/10.1021/acsomega.2c04765>.



The Prins reaction.

⁵ "The Catalytic Asymmetric Intermolecular Prins Reaction." C.D. Díaz-Oviedo, R. Maji, B. List, *J. Am. Chem. Soc.*, 2021, 143, 20598–20604. <https://doi.org/10.1021/jacs.1c10245>.

List's paper included statements that the Prins reaction is used in the enzyme-catalyzed cascade cyclization of terpenes in natural products biosynthesis.⁶ This led me to a paper from Dean Tantillo at UC Davis that analyzed the mechanism of this cyclization using computational chemistry.^{7,8} In that paper, the authors describe a delocalized 2-electron-3-centre carbocation described as a *nonclassical cation*.

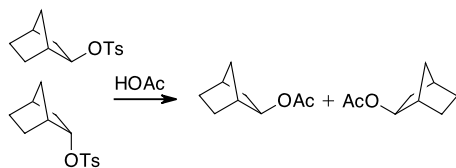
If you search the phrase “nonclassical cation”, you will see the results dominated with commentary on the great war of physical organic chemistry. I dropped my line of inquiry on the Prins reaction and began exploring the trenches of the nonclassical ion conflict.

The Nonclassical Cation

Reactions in strong acid were common investigations in the early decades of organic chemistry. Boil something in sulphuric acid, analyze the products, and become famous. That is what Hendrik Prins did. Interesting things can happen in such forcing conditions.

Saul Winstein's Indecent Proposal

In 1949, Saul Winstein,⁹ working at UCLA, reported results for the solvolysis of tosylates of norborneol in acetic acid that indicated a seemingly unlikely racemic product. Further work showed that both the chiral *exo* and *endo*-norbornyl tosylates gave only the racemic *exo*-norbornyl acetate product as shown in Figure 1. Furthermore, the *exo* starting material reacted much faster, indicating electronic participation by the adjacent σ -bond. The best explanation that Winstein could offer was to propose a strange intermediate that contain partial bonds.^{10,11,12}



The Classical Mechanism

This reaction may seem mysterious, but it can be interpreted as both starting materials giving the same cationic intermediate. The *exo* face of the intermediate is more available for the addition of the nucleophile and this may explain the preponderance of the *exo* product. The racemization could be the result of the isomerization, via a Wagner-Meerwein rearrangement, to give an identical cation whose product is the enantiomer. Figure 2 on the following page shows this classical interpretation.

⁶ If you have taken the natural products course, you will be familiar with this idea.

⁷ “Theoretical Studies on Farnesyl Cation Cyclization: Pathways to Pentalenene.” P. Gutta, D.J. Tantillo, *J. Am. Chem. Soc.*, 2006, 128, 6172–6179. <https://doi.org/10.1021/ja058031n>.

⁸ If you are taking the computational chemistry course, then you may be better able to interpret the methods in this paper than I am.

⁹ Saul Winstein was born in Montreal in 1912; he was Canadian. His family moved to California when he was 11 years old.

¹⁰ I found dozens of reviews that explore this subject. Here is one to start you down your own rabbit-hole: “Norbornyl Cation Isomers Still Fascinate.” P.v.R. Schleyer; V.V. Mainz; E.T. Strom, In *The Foundations of Physical Organic Chemistry: Fifty Years of the James Flack Norris Award*, American Chemical Society Symposium Series, 2015, Vol. 1209, 139–168. <https://doi.org/10.1021/bk-2015-1209.ch007>.

¹¹ Here is another review that includes a discussion on how the nonclassical ion debate was an example of scientific disagreement done poorly. It got personal. “Sustaining a Controversy: The Non-classical Ion Debate.” W. Goodwin, *The British Journal for the Philosophy of Science*, 2013, 64, 787–816. <https://doi.org/10.1093/bjps/axs025>

Figure 1: Solvolysis of norbornyl tosylates in acetic acid.

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¹² EXPLORATION TIP: I used these reviews to guide me to the original literature. Observe that, in this exploration, we will quickly leave these broad reviews behind and focus on a few sets of data from the primary literature. Reviews are useful, but the purpose of these explorations is to dig into and appreciate some experimental data and analysis.

For a 3D interactive image of a 2-norbornyl structure see the PubChem database for 2-chloronorbornane at <https://pubchem.ncbi.nlm.nih.gov/compound/12220913>.

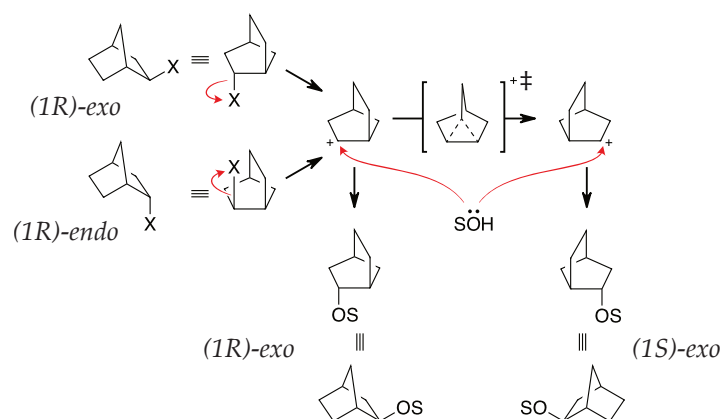


Figure 2: Classical mechanism for solvolysis of norbornyl tosylate.

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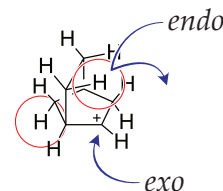


Figure 3: Addition to the carbocation is more accessible at the *exo* face of the intermediate. ↑

The Nonclassical Mechanism

The classical mechanism presented in Figure 2 may explain the results of the product analysis¹³ but it does not explain the rate observations reported by Winstein. Both the *endo* and the *exo* starting materials lead to the same secondary carbocation intermediate, yet the *exo* reactant has a rate that is 350-fold greater than the *endo* reactant.

¹³ This is one of the methods described in chapter 8.8. See chapter 8.8.1 and 8.8.6 of the textbook.

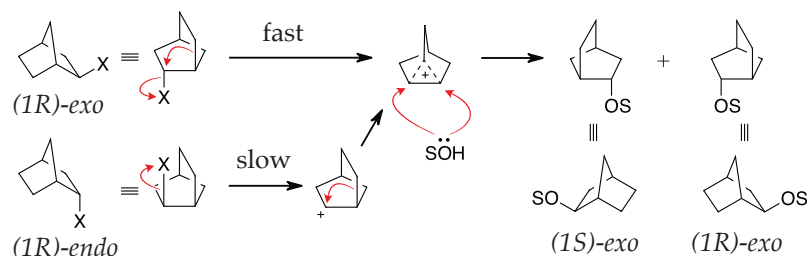


Figure 4: Nonclassical mechanism for solvolysis of norbornyl tosylate.

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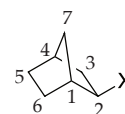


Figure 5: Numbering for norbanes according to the IUPAC rules for bicyclic compounds. ↑

In the nonclassical mechanism presented in Figure 4, the departure of the leaving group in the *exo* case is assisted by electron donation from the σ -bond. Winstein proposed that this provided a route to the nonclassical intermediate, which is symmetrical and solvent can add at either corner to give the racemic *exo* product. The *endo* starting material must form a secondary cation without this assistance and then the σ -bond can stabilize structure. As a result, the formation of the intermediate nonclassical carbocation is much faster with the *exo* system.

The Declaration of War

Winstein's report excited many organic chemists. The idea of these non-classical ions had been discussed as early as 1939, but it was Winstein's work

that provided the first rigorous evidence. Many others began investigating this phenomenon and the term "nonclassical" ion was coined.

We are comfortable writing resonance structures and dotted lines these days, but in 1949, the idea of partial bonds was controversial. The comfortable solid line of the Lewis structure was the standard, and graphical conventions for representing other electronic arrangements were yet to be agreed upon. The chemists working on nonclassical cations enthusiastically took artistic license. Many dead ends and poorly designed experiments clouded the issue. But evidence of the nonclassical intermediate began to mount.

H.C. Brown was well on his way to his 1979 Nobel prize and had had enough of this craziness. He laid down the law in 1962 with a series of papers attacking the idea of the nonclassical ion and proposing an alternate hypothesis. In Brown's opinion, the rate differences were due to steric effects in the transition state for formation of the initial secondary cation, and the racemization was the result of rapid isomerization of the norbornane backbone due to well-understood carbocation rearrangement mechanisms. There was no need to propose the existence of a cryptic intermediate that no one seemed to be able to catch.¹⁴ Most chemists were on team Winstein. Still, Brown and his enormous intellectual empire of grad students and postdoctoral fellows waged a battle fought with robust, well-reasoned arguments, solid experimental evidence, snide remarks and outright vitriol at conferences.

In 1986, the writing was on the wall, and the idea of the nonclassical cationic intermediate had been clarified, better defined and well supported by evidence. H.C. Brown declared he wasn't going to waste any more time arguing, but never admitted defeat.¹⁵ Winstein had died in the height of his career in 1969, so perhaps the fun of arguing with a worthy opponent was gone. Or, maybe it was George Olah's NMR observations of the cation, proving it existed as an intermediate in the time frame of an NMR experiment.

The Evidence

I will now examine some selected results used in this intellectual conflict. First, I will present an overview of Winstein's and others' kinetic results that supported the hypothesis for the nonclassical norbornyl cation. Then, I will present a Hammett plot by H.C. Brown that supports the classical ion. Finally, I will explore some isotope scrambling results from John Roberts that support the nonclassical view.

In the end, I will briefly describe the NMR results of Olah and the X-ray crystal structure of the norbornyl cation (a Nobel prize-worthy effort) observed by Meyer and Crossing in 2013 that proved that the nonclassical structure exists, at least in that one case. And, to prove that it all still matters, I will briefly introduce the 2024 computational study by Houk that showed a kinetic pathway to a stable nonclassical norbornyl cation at 25 °C.¹⁶

EXPLORATION TIP: I have referenced the textbook to show where it contains topics related to this exploration. The textbook presents a thorough discussion of Winstein and Brown's disagreements. It was a major topic in the field for two decades and, as you shall see, was still the subject of Nobel-class work in this century. I would be surprised if your topic were not mentioned in some way in the textbook, and you should have included textbook references in your exploration. The textbook is a valuable resource. See Chapter 11.5 for more information on the S_N1 mechanism and all its wrinkles. Use the textbook index to find commentary that will help with your work.

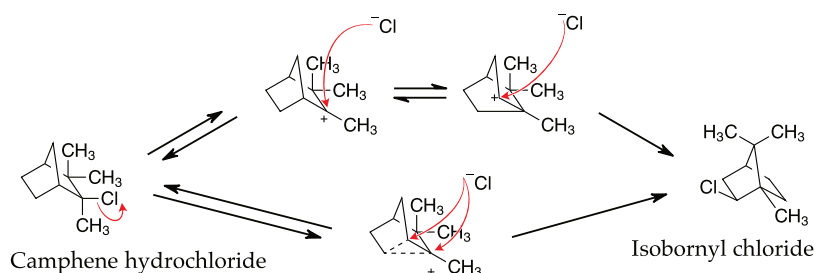
¹⁴ Trapping of an intermediate is another of the methods described in chapter 8.8. See chapter 8.8.3 of the textbook

¹⁵ H.C. Brown was awarded the 1979 Nobel prize for chemistry in recognition of his work with boranes. George Olah was awarded the 1994 Nobel Prize for Chemistry for his work on carbocations. Winstein, had he lived long enough, would surely have shared in that prize or been awarded a prize earlier.

¹⁶ EXPLORATION TIP: As you read this document, you will see that I did not get past the first goal. There was much more to explore in the Winstein work than I expected, and I ended up with 15 pages on that topic alone. I should now rewrite this text to reflect what I actually did as if it were the original plan all along. I could also drop the story about starting with the Prins reaction. I'll keep the history of the conflict between Brown and Winstein. I think that story makes this report more interesting. How would you improve this document? A good editor will make writing shorter and better.

Wilson's Observation

The idea for the nonclassical cation was not Winstein's idea. He was using norbornane as a model to investigate Christopher Wilson's report of nonclassical ions in rearrangements of camphene in hydrochloric acid.¹⁷



This reaction could easily be explained by the dissociation of the chloride leaving group to give the tertiary cation product. The subsequent Wagner-Meerwein rearrangement to produce the secondary cation intermediate that gives the isobornyl product is possible, but generally these rearrangements move toward the more stable cation, not away.¹⁸ The reaction was much faster than the typical dissociation of tertiary chlorides. Wilson proposed that the 1,6- σ -bond participated in forming the cationic intermediate via the nonclassical cation to explain the high rate of the reaction.

Winstein's Observations

Winstein investigated the solvolysis of 2-norbornyl tosylates and brosylates to test this hypothesis. He chose the norbornyl system to keep things simple and eliminate other possible effects in cation formation that could explain the results. He planned the experiment to show that the cation formation of the *exo* system would be much faster than in the case of the *endo* system (see Figure 4 on page 3). Winstein reported his results in a one-page communication in 1949 and reported the detailed results two years later.^{19,20,21,22} The observations reported were as follows...

1. The *exo* stereoisomer reacted 350-fold faster than the *endo*.
2. The products of acetolysis of both *exo* and *endo*-2-norbornyl brosylate were exclusively the *exo*-2-norbornyl acetate. No *endo* product was observed.
3. The reaction of enantiomerically pure starting material resulted in a racemic *exo*-2-norbornyl acetate product mixture. The rate of disappearance of the *endo* starting material matches the rate of loss of optical activity but, in the *exo* case, the rate of loss of optical activity is much greater than the disappearance of reactant.

¹⁷ There are many review articles that describe the story of Winstein and Brown. The early hints of the nonclassical ion are described in "What's in a Name?" From Designation to Denunciation - The Nonclassical Cation Controversy." S.J. Weininger, *Bull. Hist. Chem.*, 2000, 25, 123-131. <https://www.ideals.illinois.edu/items/133259>

Figure 6: Rearrangement of camphene hydrochloride as reported by Christopher Wilson.

¹⁸ Chapter 11.5.12 of the textbook discusses Wagner-Meerwein rearrangements.

¹⁹ "The Structure of the Bicyclo[2,2,1]2-Heptyl (Norbornyl) Carbonium Ion." S. Winstein, D.S. Trifan, *J. Am. Chem. Soc.*, 1949, 71, 2953. <https://doi.org/10.1021/ja01176a536>

²⁰ "Neighboring Carbon and Hydrogen. X. Solvolysis of *endo*-Norbornyl Arylsulfonates." S. Winstein, D. Trifan, *J. Am. Chem. Soc.*, 1952, 74, 1147-1154. <https://doi.org/10.1021/ja01125a006>

²¹ "Neighboring Carbon and Hydrogen. XI. Solvolysis of *exo*-Norbornyl p-Bromobenzenesulfonate." S. Winstein, D. Trifan, *J. Am. Chem. Soc.*, 1952, 74, 1154-1160. <https://doi.org/10.1021/ja01125a007>

²² Parts VI through XI of Winstein's investigations into the effect of "neighboring carbon and hydrogen" groups on the formation of intermediates in solvolysis reactions were reported as a group of papers in the 5th issue of JACS in 1952. The first 44 pages of this journal issue were dedicated to eight papers from Winstein's group.

4. In the case of the *endo*-2-norbornyl brosylate, the *exo* product retained a small amount of optical activity while the *exo*-2-norbornyl brosylate produced a completely racemic *exo* product.

Winstein's Conclusions

Winstein published his evidence in two consecutive papers in JACS: one for the *endo* case²⁰ and the other for the *exo* case.²¹ You will need to read both of them to assemble the four sets of evidence stated above.

The observations support the hypothesis of a nonclassical carbocation as an intermediate in the mechanism. Below are Winsteins conclusions based on the observations. The possible mechanisms the Winstein considered are presented in Figure 7 on the following page.

1. In the *exo* case, the σ -6,2-bond can donate into the antibonding orbital of the bond to the leaving group and assist in its departure. This leads to rapid formation of the nonclassical intermediate. This intermediate is more stable than a typical secondary carbocation due to the distributed charge.²³ In the *endo* case, this assistance is not possible and the secondary carbocation must form first before it rapidly collapses into the same nonclassical intermediate. Without the anchimeric assistance of the σ -6,2-bond the rate of solvolysis for the *exo* case should be similar to the *endo* case.²⁴
2. The lack of observation of any *endo* product indicates that a traditional secondary cation intermediate is unlikely to exist. Such an intermediate could be attacked at either face. While the *exo* product is more likely, the *endo* product should be observed in measurable amounts.
3. The racemic product can be explained by the classical mechanism (see Figure 2 on page 3). However, in the *exo* case, the mismatch in the rate as measured optically or by titration supports the idea of rapid formation of a nonclassical intermediate that can react with the departing leaving group to return to either enantiomer of the starting material (see Figure 4 on page 3). The rate of separation of the ion pair to allow solvent attack is much slower than the rate of return to a racemic starting material.²⁵ The fact that this difference in rate is not observed in the *endo* case further supports the hypothesis of the σ -6,2-bond providing anchimeric assistance for the formation of the nonclassical intermediate.²⁶
4. There is a small percentage of the solvolysis reaction with the *endo* starting material that gives the stereospecific product from an S_N2 mechanism.²⁷ The product mixture is therefore not perfectly racemic and retains a small amount of optical activity. The *endo* case shows no competing S_N2 mechanism.

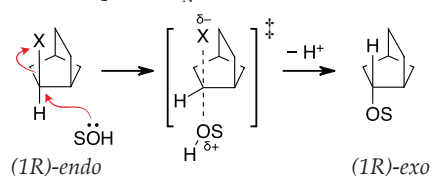
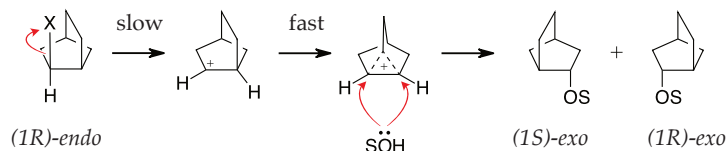
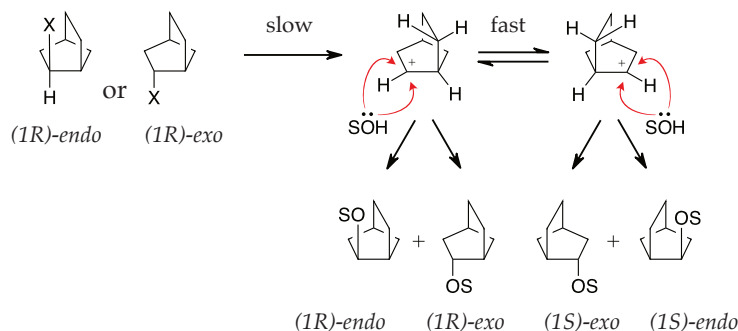
²³ Chapter 2.2.3 of the textbook discusses the stability of various carbocations and presents Table 2.8 on page 88, which contains values for hydride-ion affinity measured for various carbocations.

²⁴ Chapter 11.5.14 discusses S_N1 reactions where nonclassical mechanisms are observed or suspected and Chapter 14.5.5 presents a discussion on the electronic structure of nonclassical carbocations using molecular orbital theory.

²⁵ Chapter 11.5.8 discusses the idea of an initial ion pair being an intermediate in the dissociation of the leaving group in S_N1 mechanisms.

²⁶ Chapter 11.5.13 presents the concept of anchimeric assistance in carbocation chemistry.

²⁷ It may not be a true S_N2 mechanism, but may be an example of a nucleophilic solvent assisting the formation S_N1 intermediate and then immediately adding, resulting in inversion. See Chapter 11.5.6 for a discussion on the concept of nucleophilic assistance by solvent.

1) Stereospecific S_N2 substitution of solvent-assisted S_N1 2) Formation of 2° cation followed by conversion to nonclassical intermediate3) Formation of 2° cation followed by rapid Wagner-Meerwein rearrangements

4) Rapid formation of nonclassical intermediate as ion pair

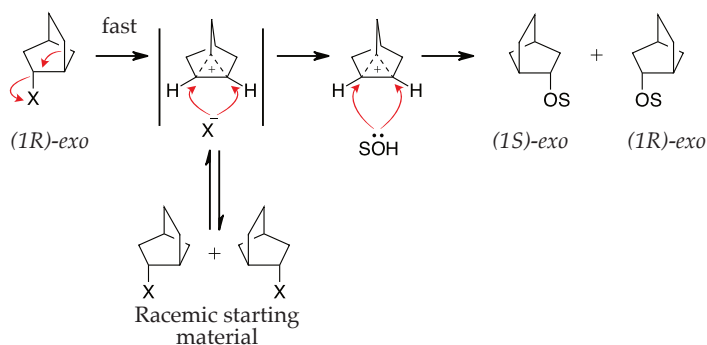


Figure 7: Possible mechanisms considered by Winstein for the solvolysis of *endo* and *exo*-2-norbornyl brosylates

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With these conclusions in mind, Winstein argues that the *endo*-norbornyl starting material can either undergo a slow S_N2 substitution with solvent or a slow dissociation to give the secondary carbocation intermediate, which quickly rearranges to become the nonclassical cation that produces the racemic *exo* product mixture (pathways 1 and 2 in Figure 7). This hypothesis explains the observation that the rate of loss of optical activity closely matches

the rate of product appearance and it explains why the optical activity did not completely vanish.

The *exo*-norbornyl starting material can proceed via rapid formation of the nonclassical intermediate. The reaction is fast and an initial ion pair within a solvent cage is produced that can react with the leaving group and return to a mixture of racemic starting material. This process is much faster than the separation of the ion pair followed by addition of the solvent nucleophile to give the racemic *exo* product mixture (pathway 4 in Figure 7). The reaction is very fast and none of the slow S_N2 pathway contributes to the product mixture.

The possibility of a completely classical route that explains the observations (pathway 3 in Figure 7) was discounted by Winstein because no *endo* product was observed and because the rate difference between the *endo* and *exo* reactants was too large to be consistent with both forming the same initial secondary carbocation intermediate.

The great minds of physical organic chemistry would battle about whether route 3 or route 4 in Figure 7 is the better explanation for the observation over the following two decades. We will leave that behind for now and explore the details of the experiments that Winstein and Trifan performed to provide the evidence for their revolution against classical cations.^{20,21}

Exploring the Methodology

When reading a paper, I always read the methods carefully. I learn new techniques this way and can better interpret the results with these details in mind. As I read the two papers by Winstein and Trifan,^{20,21} I found myself asking some questions.

1. How were the *exo* and *endo* starting materials synthesized and characterized?
2. How were the enantiomers resolved? Why did they use “A” and “B” rather than “R” and “S”?
3. How were the products characterized?
4. How were the rate determined by optical activity and by titration?²⁸

Hopefully the papers will provide the answers I need.

The Diels-Alder Reaction

The Diels-Alder reaction was introduced in 1928 and was well explored by 1952. It was known to be stereoselective and chemists could predictably use it to make products with a majority of the *endo* stereoisomer. The reason for this stereoselectivity was explained by Woodward and Hoffman in 1965.^{29,30,31}

²⁸ I was especially curious about how the authors titrated in pure acetic acid.

²⁹ “Evolution of the Diels-Alder Reaction Mechanism since the 1930s: Woodward, Houk with Woodward, and the Influence of Computational Chemistry on Understanding Cycloadditions.” K.N. Houk, F. Liu, Z. Yang, J.I. Seeman, *Angew. Chem. Int. Ed.*, 2021, 60, 12660–12681. <https://doi.org/10.1002/anie.202001654>.

³⁰ Otto Diels and Kurt Alder were awarded the 1950 Nobel prize for chemistry. Roald Hoffman was awarded the Nobel prize for chemistry in 1981. Alas, Robert Woodward had passed away in 1979.

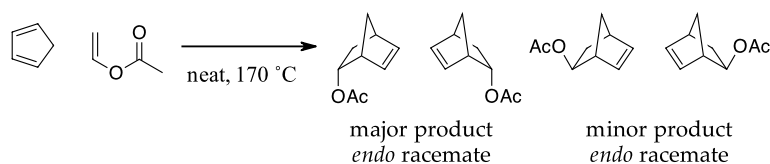
The UPEI Chemistry Department has a connection to this famous reaction. Cirill Schmidt (1930–2024) worked with Robert Woodward at Harvard before being recruited by Regis Duffy to come to St. Dunstan’s University. His early work at UPEI reflected this background. “The transition state of Diels-Alder reactions.” M.T.H. Liu, C. Schmidt, *Tetrahedron*, 1971, 27, 5289–5297. [https://doi.org/10.1016/S0040-4020\(01\)91693-X](https://doi.org/10.1016/S0040-4020(01)91693-X).

³¹ Chapter 15 of the textbook presents the Diels-Alder reaction in detail.

Winstein and Trifan used the Diels–Alder reaction to make *endo*-norbornyl acetate from cyclopentadiene and vinyl acetate. The IR spectrum of *exo* and *endo*-norbornyl acetate was known and the assignment of *endo* and *exo* in the norbornyl system had been established previously by Kurt Alder himself. It was known that catalytically reducing norcamphor will stereoselectively produce *endo*-norborneol. By comparing esters of this synthetic norborneol with corresponding esters of the natural product camphor, Alder was able to establish the stereochemistry.^{32,33}

Synthesis of *endo*-Norbornyl Acetate

The synthesis of *endo*-norbornyl acetate by Winstein and Trifan is reported in the first of the two papers.²⁰ They reacted a large amount of neat cyclopentadiene and vinyl acetate at 170 °C to obtain a racemic mixture that was mostly the *endo* stereoisomer. They collected the product by vacuum distillation.



³² “Zur Kenntnis der Dien-synthese. XI. Über die Dien-synthese mit Vinylestern und halogenierten Äthylenen. Einfache Wege in die Norcampher-reihe.” K. Alder, H.F. Rickert, *Justus Liebigs Ann. Chem.*, 1940, 543, 1–27. <https://doi.org/10.1002/jlac.19405430102>.

³³ NOTE: Always read papers that you reference, no matter what language. The above paper³² is in German, like most important publications in chemistry before 1939. I had Google translate do a poor job of translating it, and I believe that I understand how the stereochemistry of synthetic norborneol derivatives was established in this case.

Figure 8: Stereoselective Diels–Alder synthesis of *endo*-norbornyl acetate
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Purification of Racemic *endo*-Norbornyl Acetate

The authors measured reaction rates using racemic mixtures of *endo*-2-norbornyl reactants. The rate of change in optical rotation and the rate measured by titration were nearly identical in these cases, so they did not need to investigate the reaction using pure enantiomers. To separate the *endo*-product from the *exo* impurities, they converted the norbornyl acetate to the phthalate ester and used crystallization to obtain the pure racemic *endo*-norbornyl phthalate. They recrystallized the product six times to obtain pure product. The phthalate was characterized by comparing the melting point to that reported by Alder.³²

To obtain a pure sample of *endo*-2-norbornyl acetate, the authors hydrolyzed the phthalate group and esterified the *endo*-2-norborneol product to obtain pure *endo*-2-norbornyl acetate. The IR spectrum reported in the paper²⁰ was made using this material (more on that later). I took note of the liberal use of benzene as a solvent. It was the good old days.

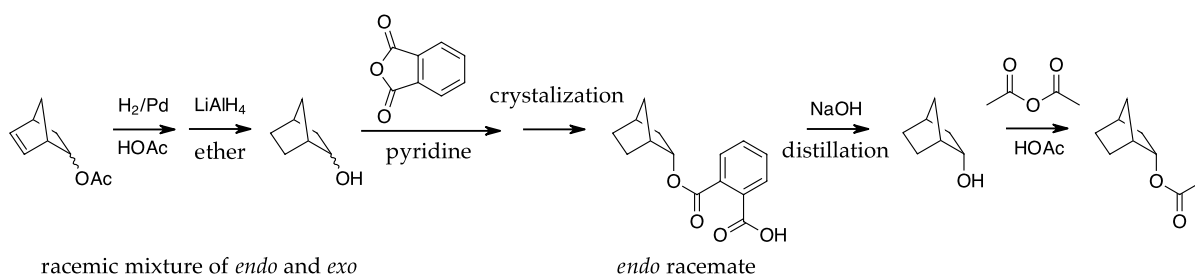
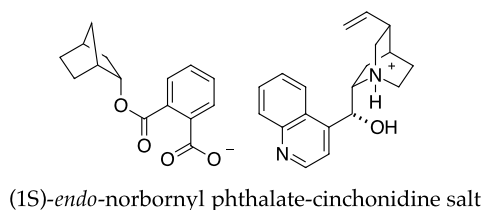
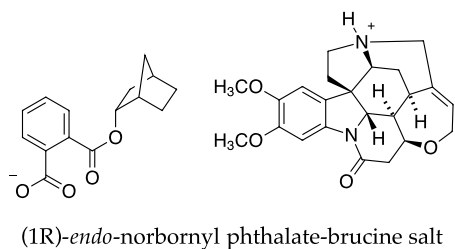


Figure 9: Purification of racemic *endo*-norbornyl acetate via the phthalate. ↑

Resolution of (1*R*) and (1*S*)-*endo*-Norbornyl Acetate

The authors used co-crystallization with alkaloid natural products to obtain enantiomerically pure samples of what they named as the “A and B enantiomorphs”. The establishment of a standardized nomenclature for stereochemistry was still twenty years in the future.^{34,35} I looked up the optical rotation of *endo*-norborneol and found that the (+)-enantiomer is (1*S*)-*endo*-norborneol.^{36,37} The authors reported a positive rotation for their “B enantiomorph”. I believe that I can now say that the “A” is (1*R*)-*endo*-norborneol and the “B” is (1*S*)-*endo*-norborneol. I will use the CIP designations in this report, as IUPAC intended.



The authors mixed warm acetone solutions of the racemic *endo*-norbornyl phthalate and brucine.³⁸ The precipitate was found to contain primarily the (1*R*)-*endo*-norbornyl phthalate. The mother liquor was then mixed with a solution of cinchonidine and the precipitate thus obtained was found to be mostly the (1*S*)-*endo*-norbornyl phthalate. The norbornyl phthalates were separated from their salts by acidification and extraction into benzene (so much benzene was used in this work, it's scary). After multiple recrystallization steps, pure (1*S*) and (1*R*)-*endo*-norbornyl phthalate was obtained.

³⁴ “The 50th Anniversary of the Cahn–Ingold–Prelog Specification of Molecular Chirality.” G. Helmchen, *Angew. Chem. Int. Ed.*, **2016**, *55*, 6798–6799. <https://doi.org/10.1002/anie.201603313>.

³⁵ Chapter 6.1 of the textbook presents the rules for stereochemical designations.

³⁶ <https://www.sigmaaldrich.com/CA/en/product/aldrich/543519>. Accessed August 6, 2025.

³⁷ NOTE: I disapprove of websites as references. They change and links break. Here I have used a reference to the Sigma-Aldrich catalog. Next week, this link might not be there. However, it was the only reference that I could find in five minutes of searching. I found papers that described using the optical rotation of norbornyl esters in characterization, but they then referenced other papers for the actual data. I didn't feel like going down those rabbit holes. I have data from Sigma-Aldrich and that is good enough. Judge me, blame me, but never change me.

Figure 10: Brucine and cinchonidine salts of *endo*-norbornyl phthalate enantiomers. ←

³⁸ Chapter 5.8 of McMurry's Organic Chemistry, 10th ed. discusses the resolution of enantiomers. It is available at <https://openstax.org/details/books/organic-chemistry>

Synthesis of Sulfonate Esters

The phthalate esters were converted to the corresponding norborneols by hydrolysis. Then these were converted to either the *p*-toluenesulfonate ester (tosylate) or the *p*-bromobenzenesulfonate ester (brosylate) via the corresponding sulfonyl chloride.

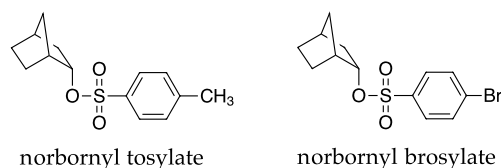


Figure 11: Reactants for solvolysis study

Synthesis of the *exo*-Norbornyl Materials

The second paper²¹ is dedicated to the *exo*-norbornyl system. The synthesis of *exo*-2-norborneol is simply the hydrolysis of *endo*-2-norbornyl tosylate. The reaction, as discussed above, is observed to produce only the *exo* product. The racemic *exo*-2-norborneol is purified by distillation and then converted to the phthalate. This can then be used to synthesize the racemic *exo*-2-norbornyl acetate, tosylate or brosylate as outlined above. the *exo*-2-norbornyl acetate was used to produce the IR spectrum discussed later.

The resolution of the enantiomers is achieved by co-crystallization of the racemic phthalate with cinchonidine in acetone to produce a precipitate enriched in the (1*S*)-*exo*-norbornyl phthalate. The mother liquor was thus enriched in the (1*R*)-*exo*-norbornyl phthalate. Using these two enriched sources and many recrystallizations and extractions, samples of purified enantiomers were obtained and converted to the brosylate as described above.

Characterization by IR

After the synthesis and purification steps described above, the authors now possessed samples of *exo*-norbornyl-2-acetate and *endo*-norbornyl-2-acetate. In the first figure in each of the two papers^{20,21} they present the IR of each.³⁹ I digitized the spectra data and plotted the spectra using modern *Python* tools. I believe that this is an improved presentation of the IR data and better highlights how the two stereoisomers can be identified by IR. See Figures 12 and 13 on the next page.

The fact that the *exo*-norbornyl reactants always give *exo* products and the *endo* reactants also give *exo* products is proved by isolating the acetic acid solvolysis product and taking the IR.

³⁹ The IR spectra appear to be hand drawn from data. The authors present the spectra from a neat film and from a CCl₄ cell. Confusingly, they use a dotted line for the neat film in one paper and, in the other paper, a solid line is used.

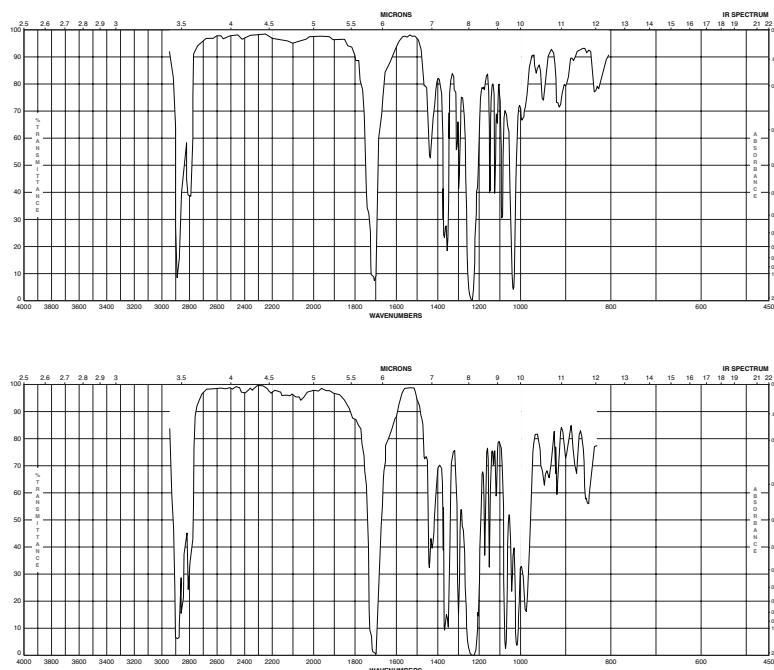


Figure 12: IR spectra of *endo*-2-norbornyl acetate (top) and *exo*-2-norbornyl acetate (bottom) in CCl_4 .

The instrument is described as a Baird infrared spectrometer. This is a forgotten brand first produced by Baird Associates in 1945. It is likely a dispersive double beam instrument and would have been state of the art at the time. William Baird founded the company in 1936 and ran it until his death in 1982. Like many scientists at the time, his spectrometer and other inventions were developed as part of the Manhattan Project (although he would never have known what his devices were being used for at the time). You can learn more by reading “The History of Baird Corporation: A Broad Perspective on the Progress of Industrial Spectroscopy.” D.F. Walsh, *Applied Spectroscopy*, 1988, 42, 1336–1350. <https://doi.org/10.1366/0003702884429607>

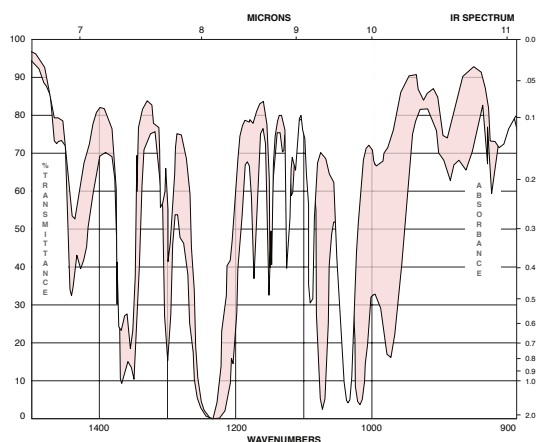


Figure 13: Fingerprint region of IR spectra of *endo*-2-norbornyl acetate (front) and *exo*-2-norbornyl acetate (pink, back) in CCl_4 .

Following the Solvolysis Reaction with Titration

The authors reacted the norbornyl sulfonates in pure acetic acid, 75 % acetone-water and ethanol. I will discuss only the acetolysis results in detail.

The authors followed the reaction by titration. For the *endo* stereoisomer the solvolysis reaction was very slow at room temperature. The reaction was performed at 75 °C and aliquotes were taken out at time points and quickly cooled to 25 °C for analysis.²⁰ The authors consider any reaction progress that occurs during the analysis to be insignificant. For the *exo* stereoisomer,

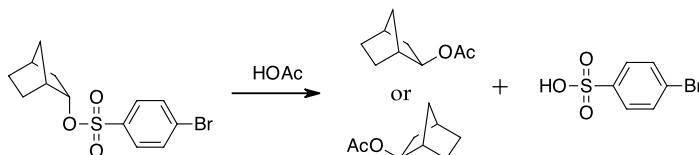
the solvolysis was carried out at 25 °C. Aliquots taken were quickly diluted with petroleum ether to slow the reaction for analysis. The authors state that this did not alter the results but provide no proof in the paper.²¹

The titration method was not described in the methods of the two papers that we are exploring but was referenced from a previous investigation from Winstein's group.^{40,41,42} They use a traditional titration with a colourimetric indicator, bromophenol blue.

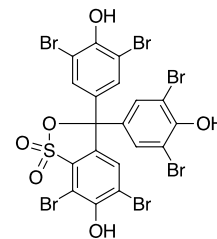
Bromophenol blue is yellow below pH 4 in water and blue at higher pH. The authors report that there is a second transition is strong acid where the dye transitions to colourless.⁴³ The product of the solvolysis of a sulfonate ester is sulfonic acid, a very strong acid. The authors report that Bromophenol blue is yellow in acetic acid if even a trace of acetate ion is present and is colourless when stronger acid is present. So the authors titrate the sulfonic acid/acetic acid product mixture with a sodium acetate/acetic acid solution. The strong sulfonic acid will transfer protons to the acetate until it is consumed. The next drop of acetate solution will switch the colour to yellow. Thus, the authors can titrate with a high degree of accuracy for the amount of sulfonic acid product produced.

Following the Solvolysis Reaction with Optical Activity

In addition to the sulphonic acid, the reaction produces a racemic mixture of acetate ester product. If we start with a pure enantiomer we will be able to measure the optical activity and watch it reduce to zero as the reaction proceeds.



The authors measured the optical activity using a water-jacketed polarimeter. The reaction mixture was placed in the polarimeter tube and readings were taken over time.⁴⁴



Bromophenol blue

⁴⁰ "The Role of Neighboring Groups in Replacement Reactions. XII. Rates of Acetolysis of 2-Substituted Cyclohexyl Benzenesulfonates." S. Winstein, E. Grunwald, L.L. Ingraham, *J. Am. Chem. Soc.*, 1948, 70, 821–828. <https://doi.org/10.1021/ja01182a113>

⁴¹ Again, in this issue of JACS, we see Winstein publish a group of papers. Here, he presented parts X through XIV of his series investigating "the role of neighboring groups in replacement reactions."

⁴² Interestingly, at the end of that batch of papers is the seminar paper that establishes the Grunwald-Winstein equation for measuring the ionizing power of solvents: "The Correlation of Solvolysis Rates." E. Grunwald, S. Winstein, *J. Am. Chem. Soc.*, 1948, 70, 846–854. <https://doi.org/10.1021/ja01182a117>

Figure 14: The acetolysis of norbornyl-2-bromobenzenesulfonate produces a racemic mixture of exo-norbornyl-2-acetate and p-bromobenzenesulfonic acid.

←

⁴³ Your favourite indicator, phenolphthalein, has a similar third transition is strong acid. It becomes orange in colour.

⁴⁴ The instrument is described as a Hilger polarimeter. You can find it in the 1928 Hilger catalog available via the Smithsonian Books Collection in the Internet Archive: <https://archive.org/details/Generalcatalogu00Hilg>. Accessed Aug. 7, 2025

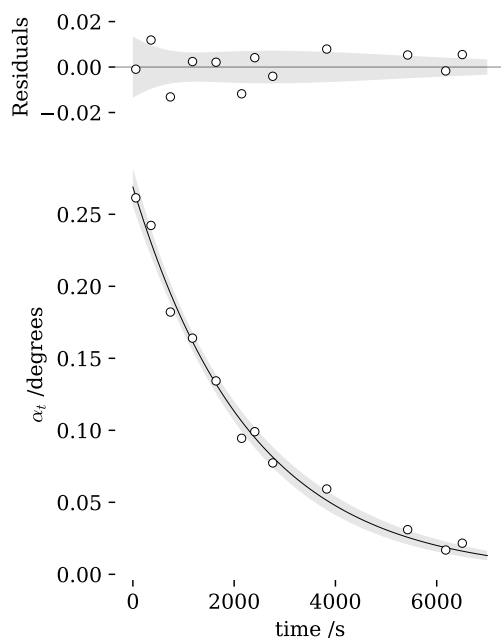
Testing the Numbers

We have explored Winstein's observations and methods. Now let us use the data available to see if we can obtain the same results using modern data analysis.

Reporting Rate Data by Optical Activity

Data is sparse here, as in most vintage papers, but we have a plot of optical activity over time for solvolysis of (Figure 2 in reference 21 as reproduced in Figure 15 here.). I digitized the plot using *WebPlotDigitizer* and fit the data to the first order rate law. You may have noticed that the authors presented the plot and rate law differently than you were trained to do.

We would set up the integrated first-order rate law as $\ln \frac{\alpha_t}{\alpha_0} = -kt$ where α is the optical rotation. I will not present that linear equation, but will use the form of $\frac{\alpha_t}{\alpha_0} = e^{-kt}$ in my own data analysis because non-linear plots are easy to optimize using *Python* tools.



The curve fit is presented in Figure 16 and it agreed well with the reported data from Table 1 of reference 21 ($\alpha_0 = 0.265^\circ$ and $k_{obs} = 4.25 \times 10^{-4} \text{s}^{-1}$).

350-Fold?

The authors state that the *exo*-norbornyl-2-bromobenzenesulfonate reacts 350 times faster in acetic acid compared to the *exo* stereoisomer. But the data presented in the two papers^{20,21} is at different temperatures and not directly

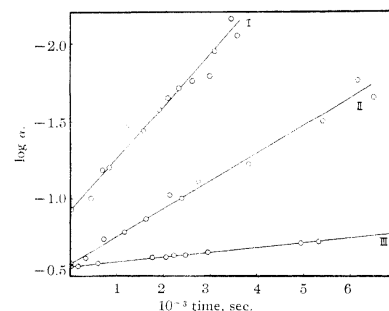


Fig. 2.—Polarimetric observation of solvolysis of *exo*-norbornyl *p*-bromobenzenesulfonate: I, in 75% acetone; II, in AcOH; III, in EtOH.

Figure 15: The solvolysis of (1*S*)-*exo*-norbornyl-2-bromobenzenesulfonate as presented in Figure 2 of reference 21. ↑

Figure 16: The acetolysis of (1*S*)-*exo*-norbornyl-2-bromobenzenesulfonate using data from Figure 2 of reference 21. ←

The confident interval is 95 %. The curve fit determined the following parameters...

$$\alpha_0 = 0.279 \pm 0.006^\circ$$

$$k_{obs} = (4.32 \pm 0.18) \times 10^{-4} \text{s}^{-1}$$

The data was plotted using a *python* notebook available via Google Colab at <https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Winstein/OpticalRot.ipynb>

EXPLORATION TIP: If you want to use *Python* for data analysis, feel free to steal my code or have an LLM system generate it. Spend as little time as possible learning *Python*. You are a chemist; you will likely only need to use *Python* in your career, not master it.

FUN EXERCISE: What if I do not assume that $\alpha_\infty = 0$? Try fitting the curve to the function...

$$\frac{\alpha_t - \alpha_\infty}{\alpha_0 - \alpha_\infty} = e^{-kt}$$

...and see how that affects the values of the parameters and the confidence intervals.

comparable. They reference a third paper where they extrapolated the rates to 25 °C using a two-point rate vs. temperature data set for each.⁴⁵

Gathering the Data

In Table 1, I will collect rate data from the three publications.^{20,21,45} Then I will use this data to compare with the observations of Winstein and his coauthors.

Reactant	Conc. /mM	Other	Method	Temp.	$k_{obs} / 10^{-5} s^{-1}$
<i>1S-endo</i> ²⁰	0.30		opt.*	74.5 °C	15.3 ± 0.8
<i>1S-endo</i> ²⁰	0.30		titr.	74.5 °C	17.0 ± 1.0
<i>rac-endo</i> ²⁰	0.30		titr.	74.5 °C	17.1 ± 1.0
<i>1S-endo</i> ²⁰	0.30	0.41 M KOAc	opt.*	74.5 °C	20.7 ± 0.8
<i>1S-endo</i> ²⁰	0.30	0.41 M KOAc	titr.	74.5 °C	18.7 ± 0.6
<i>rac-endo</i> ⁴⁵	0.02		titr.	74.5 °C	15.4 ± 0.2
<i>rac-endo</i> ⁴⁵	0.02		titr.	50.0 °C	0.816 ± 0.006
<i>rac-exo</i> ⁴⁵	0.02		titr.	33.7 °C	30.4 ± 0.1
<i>rac-exo</i> ⁴⁵	0.02		titr.	25.0 °C	8.79 ± 0.09
<i>rac-exo</i> ²¹	0.037		titr.	25.0 °C	8.64 ± 0.08
<i>rac-exo</i> ²¹	0.037		titr.	25.0 °C	8.90 ± 0.01
<i>1S-exo</i> ²¹	0.20	0.25 M KOAc	opt.**	25.0 °C	42.5
<i>1S-exo</i> ²¹	0.20	0.25 M KOAc	titr.	25.0 °C	12.2 ± 0.3

⁴⁵ "Neighboring Carbon and Hydrogen. VII. Reactivity of Some Alicyclic and Bicyclic Derivatives 1,2,3." S. Winstein, B.K. Morse, E. Grunwald, H.W. Jones, J. Corse, D. Trifan, H. Marshall, *J. Am. Chem. Soc.*, 1952, 74, 1127–1132. <https://doi.org/10.1021/ja01125a003>.

Table 1: Rates of acetolysis of 2-norbornyl brosylates as reported by Winstein et al.^{20,21,45}

* Winstein reports that the optical rotation of the (*1S*)-*endo* goes from an initial value of 1.26° to zero in the absence of acetate ion and from 1.20° to 0.04° in the presence of acetate ion, indicating some amount of inversion.

** The optical rotation of the (*1S*)-*exo* goes from an initial value of 0.265° to zero.

Optical vs. Titration Rates

As presented above, the authors reported that the rates, as measured by optical activity and titration, were the same for the (*1S*)-*endo* stereoisomer and differed significantly for the (*1S*)-*exo* stereoisomer. With the *exo* case, the rate measured by optical rotation was 3.5-fold greater than the rate measured by titration. The authors reported a value of 3.46 and so our data matches with the authors calculations.²¹

The presence of acetate salt was to avoid very strong acid conditions that can lead to racemization of the acetate ester products.⁴⁶ In the titrimetric analysis, the authors employed a reverse titration; titrating with a strong acid until the yellow colour of bromophenol blue faded. With this information, they can determine how much acetate remained and the difference would be the amount of sulfonic acid produced.

⁴⁶ This would be important to avoid when measuring the rate by change in optical rotation.

Extrapolating the Rates to 25 °C

To get to the 350-fold rate difference between the *exo* and *endo* reactants, the authors needed to extrapolate the rate from higher temperatures to room temperature.⁴⁵ They did not explain the method or indicate the data used but I presume that they used the two data points available for each case from Table 1. I also am guessing that they used a Eyring plot because values for $\Delta S^\ddagger$ and $\Delta H^\ddagger$ are reported.⁴⁵ Let us make the Eyring plot and see if we get anywhere near the authors reported extrapolated values at room temperature.

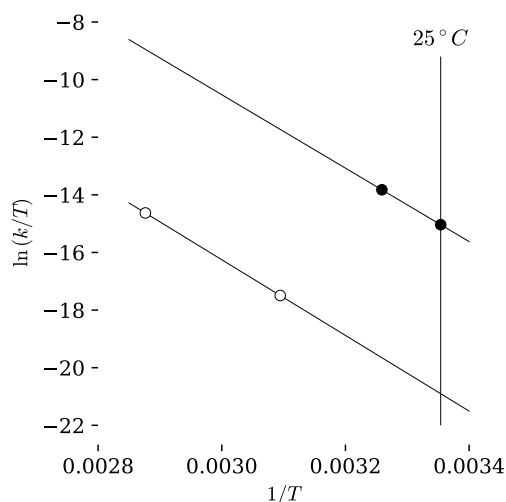


Figure 17: An Eyring plot for acetolysis of *endo* (open circle) and *exo*-norbornyl-2-brosylates (closed circles).

The data was plotted using a *python* notebook available via Google Colab at <https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Winstein/Eyring.ipynb>

Reactant	$\Delta H^\ddagger$	$\Delta S^\ddagger$	k_{obs}^{calc} at 25 °C	k_{obs}^{lit} at 25 °C
<i>rac</i> - <i>endo</i> ⁴⁵	109.2 kJ/mol	-5.0 J/mol K	$2.44 \times 10^{-7} \text{ s}^{-1}$	$2.52 \times 10^{-7} \text{ s}^{-1}$
<i>rac</i> - <i>exo</i> ⁴⁵	106.0 kJ/mol	32.9 J/mol K	$8.60 \times 10^{-5} \text{ s}^{-1}$	$8.82 \times 10^{-5} \text{ s}^{-1}$

Table 2: Eyring parameters from a classic pair of two-point line fits in Figure 17.

As you can see in table 2, the values of k_{obs} that I calculated for 25 °C and those reported by the authors were very close.⁴⁵ The thermodynamic parameters reported for the *endo* stereoisomer were also very close to the authors reported values of $\Delta H^\ddagger = 108.7 \text{ kJ/mol}$ and $\Delta S^\ddagger = -6.3 \text{ J/mol K}$

The thermodynamic parameters for the *exo* case were not reported, although the authors did state that the major difference in the kinetics was due to entropy. The values for k_{obs} featured standard deviations in both cases, indicating multiple data points were used so I do not believe that it was lack of data. The authors state that the two points were too close together for a confident determination of slope and intercept.

The large positive value for $\Delta S^\ddagger$ in the *exo* case may seem to be the opposite of what is expected for an organized transition state, such as that which leads to a bridged nonclassical carbocation. However, observe that the nonclassical cation is symmetrical and represents two degenerate structures arising from a single starting material. More equivalent options means more entropy.⁴⁷ The negative value for the *endo* case is similar to that of the solvolysis of cyclohexyl brosylate⁴⁵ and indicates a similar transition state headed toward a more traditional S_N1 intermediate.

I cannot explain why my calculations did not give exactly the same values as the authors. It was a two-point plot and the values were reported in the paper to three significant figures. I should have been spot on. I suspect that the authors used more data than they reported.⁴⁸

When inspecting the extrapolated values of k_{obs} , the ratio is 352:1 for the *exo* vs *endo* rate constants. The authors' ratio is 350:1.

⁴⁷ This was a surprisingly controversial topic. In 1988, a paper was published that presented this idea in an analysis of the kinetics of the norbornyl cation: "Ludwig Boltzmann and the norbornyl cation." G.M. Kramer, C.G. Scouten, R.V. Kastrup, E.R. Ernst, C.F. Pictroski, *J. Phys. Chem.*, 1989, 93, 6257–6260. <https://doi.org/10.1021/j100353a057>. This was followed by two letters to the editor expressing strong disagreements with the conclusions and a rebuttle by Kramer: "Response to "Ludwig Boltzmann and the norbornyl cation"." M. Saunders, *J. Phys. Chem.*, 1990, 94, 6509–6511. <https://doi.org/10.1021/j100379a065>; "Entropy and degenerate rearrangements." E. Spohr, M. Wolfsberg, *J. Phys. Chem.*, 1990, 94, 6511–6512. <https://doi.org/10.1021/j100379a066>; and "Configurational entropy of fluxional molecules." G.M. Kramer, C.G. Scouten, *J. Phys. Chem.*, 1990, 94, 6512–6514. <https://doi.org/10.1021/j100379a067>. In 1996, a paper was published that provided a thorough theoretical reasoning for the idea that symmetry equates to higher disorder. "Correlation of Entropy with Similarity and Symmetry." S. Lin, *J. Chem. Inf. Comput. Sci.*, 1996, 36, 367–376. <https://doi.org/10.1021/ci950077k>. There are so many rabbit holes to go down. I will stay out of this one.

⁴⁸ If only they had provided their calculations in a *Python* notebook, then I would be able to understand the exact method and reproduce the results precisely.

Confidence, Thermodynamics and Bootstraps

As noted above, the authors did not state thermodynamic parameters for the *exo* case. Are the two points really too close together? Is there a statistical standard for two-point plots (other than that you should never use them)? I am not a statistician, but I am sure that there is math that would allow me to estimate the confidence of the values for $\Delta H^\ddagger$ and $\Delta S^\ddagger$ using the two points provided and the reported standard errors. I do not have the time to look into this so I will use a brute-force method that is common in statistics when the math gets hard.⁴⁹

One way to estimate confidence in the parameters provided by fitting a model to a data set is to use bootstrapping.⁵⁰ This method takes a dataset and resamples it (by randomly selecting a subset of points out of a data set, fitting the model and repeating this hundreds or thousand of times. Then the 66th percentile will approximate the standard deviation or the one sigma confidence interval. Obviously, a two-point plot is entirely unsuitable for this bootstrap method. However, we have a standard error reported for each of our two points and we can use this to perform a smooth bootstrap.⁵¹ This method generates a random data value for each of the two data points based on a gaussian distribution using the reported standard error values. This can result in thousands of data sets and the collection of line-fit results will give us the median value for $\Delta H^\ddagger$ and $\Delta S^\ddagger$ as well as the confidence interval.

⁴⁹ Yes, its true – you will never use calculus again. Mostly because we can avoid by adopting dodgy methods like the bootstrap I am using here.

⁵⁰ “What Teachers Should Know About the Bootstrap: Resampling in the Undergraduate Statistics Curriculum.” T.C. Hesterberg, *The American Statistician*, 2015, 69, 371–386. <https://doi.org/10.1080/00031305.2015.1089789>.

⁵¹ We must use web references sparingly with the knowledge that they could completely change tomorrow. However, the wikipedia page for the bootstrap does give a good overview of the bootstrap method and explains why I should not be using it for a two point plot. [https://en.wikipedia.org/wiki/Bootstrapping_\(statistics\)](https://en.wikipedia.org/wiki/Bootstrapping_(statistics)). Accessed Aug. 11th, 2025. On Aug. 12th, the page might have become an homage to Horatio Alger – unless the wiki editors stop me.

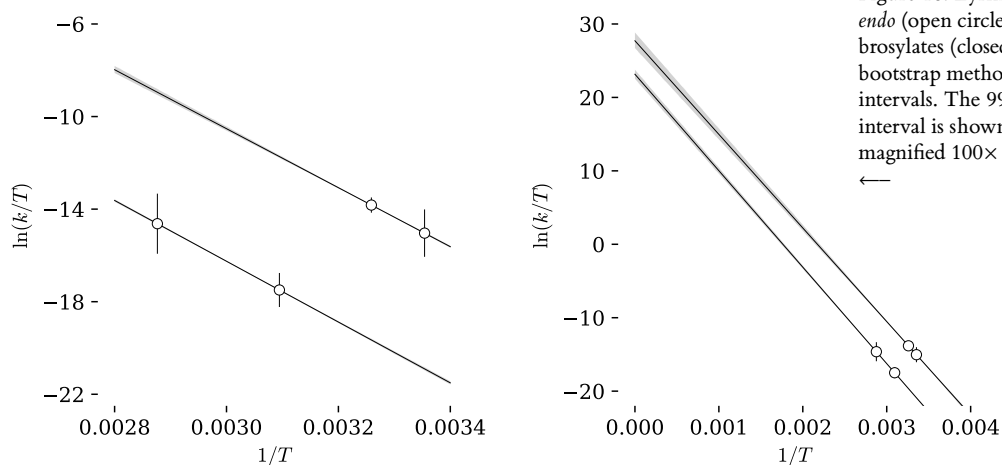


Figure 18: Eyring plot for acetolysis of *endo* (open circle) and *exo*-norbornyl-2-brosylates (closed circles) using a smooth bootstrap method to estimate confidence intervals. The 99 % (3σ) confidence interval is shown here. Error bars are magnified 100×

I won't spend any time with a further explanation. It's all in the code available on the GitHub site.⁵² If we calculate the values using a confidence interval of 66 % (one sigma) we get values as shown in Table 3 on the next page. All of these values were obtained by taking the median value of the bootstrap population (all the line fits for each randomly generated data set

⁵² The plots in Figure 18 were made using a *python* notebook available via Google Colab at <https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Winstein/Eyring.ipynb>.

Perhaps one should never do this with just two points. A statistician will be able to properly criticize my abomination because I have provided the code.

of point pairs) and the values that defines the extremes of the 33rd and 66th percentiles. The one-sigma confidence interval was very symmetric and so I just averaged it to give an estimate of the standard error for each calculated parameter.

Reactant	$\Delta H^\ddagger$	$\Delta S^\ddagger$	k_{obs}^{calc} at 25 °C
<i>rac-endo</i> ⁴⁵	(109.2 ± 0.6) kJ/mol	(−5.1 ± 1.8) J/mol K	$(2.49 \pm 0.06) \times 10^{-7} \text{ s}^{-1}$
<i>rac-exo</i> ⁴⁵	(106.0 ± 1.0) kJ/mol	(32.8 ± 3.1) J/mol K	$(8.80 \pm 0.09) \times 10^{-5} \text{ s}^{-1}$

Table 3: Eyring parameters with errors estimated using the smooth bootstrap algorithm as shown in Figure 18.

←

From the bootstrap exercise, we see that the confidence in the values of $\Delta S^\ddagger$ is acceptable. The *exo* case looks quite good for a two-point plot for which the authors hesitated to report parameters. I am skeptical – a two-point plot is never convincing. However, I believe that there is enough confidence to at least report the values, with an asterisk attached.

Summary

I explored the results in three papers published by Winstein's group in 1952 (in the same issue of JACS, back-to-back – could it be considered a single very long paper?). I summarized the observations and conclusions that the authors expressed and then explored the methods in detail. I was introduced to the methodology of following reaction rates by optical rotation and the idea of titrations in acetic acid solvent. I also enjoyed reading the synthesis details and seeing that all the reactions were easily understood at the second-year level of organic chemistry.⁵³ The resolution of the enantiomers by co-crystallization with natural products was also a good review of my organic chemistry.

I investigated the data and digitized and replotted the IR spectra. I digitized and plotted the first-order rate plot for acetolysis in a more familiar manner. The rate constant for racemization agreed well with the reported values. I plotted the two-point Eyring plots and obtained values that agreed with the author's calculations. I found no errors in the data or the data analysis.

I did not explore the confounding evidence presented by Brown or discuss how he was eventually convinced to stand aside, as I had promised. The examination of the seminal work in nonclassical cations by Winstein was more than enough to fulfill the scope of a personal exploration, and it increased my understanding of the nonclassical ion argument and the limitations faced by chemists of the 1950s.

Still on the Bleeding Edge

Was this a tempest in a teapot and long forgotten? No! It is a classic problem in physical organic chemistry and led to the development of super-acids in which carbocation intermediates are stable enough to persist. George Olah

⁵³ The Diels-Alder reaction is examined in more detail in your third-year organic chemistry course.

contributed immensely to this effort and, in 1983, he was able to obtain NMR evidence of the nonclassical norbornyl carbocation.⁵⁴ Shortly after that report, Brown left the field of battle and continued with his other very important endeavours.

The NMR evidence is convincing, but researchers wanted more than footprints; they wanted to catch this Bigfoot. In 2013, the norbornyl cation was “captured on film” when an X-ray crystal structure was obtained.⁵⁵ Both the NMR work and this X-ray experiment were performed in extremely cold temperatures, raising the possibility that the nonclassical cation is not present at room temperature. At last, in 2024, a thorough theoretical study showed a clear reaction pathway from the starting material to the nonclassical intermediate using molecular dynamics calculations.⁵⁶

The story of the norbornyl nonclassical cation is not last century – it’s last week! And maybe tomorrow?

⁵⁴ “Conclusion of the classical-nonclassical ion controversy based on the structural study of the 2-norbornyl cation.” G.A. Olah, G.K.S. Prakash, M. Saunders, *Acc. Chem. Res.*, 1983, 16, 440–448. <https://doi.org/10.1021/ar00096a003>.

⁵⁵ “Crystal Structure Determination of the Nonclassical 2-Norbornyl Cation.” F. Scholz, D. Himmel, F.W. Heinemann, P. Schleyer, K. Meyer, I. Krossing, *Science*, 2013, 341, 62–64. <https://doi.org/10.1126/science.1238849>.

⁵⁶ “Molecular Dynamics of the Norbornyl Cation in Solution and Its Generation in Winstein–Trifan Solvolysis: The Timing of Sigma Bridging.” W. Lee, T.R. Benton, A. Sengupta, K.N. Houk, *J. Org. Chem.*, 2024, 89, 1140–1146. <https://doi.org/10.1021/acs.joc.3c02325>.

Commentary on the Personal Exploration

This week, we have explored the data and arguments for and against non-classical cations. We will have gained a greater understanding of carbocation intermediates and anchimeric assistance in reaction mechanisms along the way. We won't have time to examine experimental methods and investigate the math of the papers we consider in class. That is what a personal exploration is for.

Hopefully, you have been reading the literature and have noticed a topic that could benefit from the viewpoint of physical organic chemistry. By exploring the body of work on your topic, you will soon find a paper or papers containing interesting experiments and data available to analyze.

What Goes In?

Do not use review papers or websites as your core source material. These resources are great for getting an overview of a topic and finding a body of work to explore, but you will want to dig into a small set of experiments. Find a paper that deals with an interesting problem. If it's interesting, you won't be able to stop writing. The details and data are often in previous papers referenced by the authors or in supplementary material archived by the journal. Find that data. Convince the reader that you can repeat the analysis and obtain the same conclusions as the authors. If not, find the error and propose a correction. Enjoy the feeling of finding flaws in someone else's hard work.

What Comes Out?

Consider the output captured in the exploration above. It started with an introduction that briefly told the story of how I became interested in the subject and why I selected the papers. Then I expressed my understanding of the results and conclusions according to the authors. After that, I began exploring. I went through the methods and understood how the data was obtained. I had to follow references and pull information from other sources. I examined the synthesis of starting materials. It was basic in my example, but you might find a modern and useful synthetic step that warrants a pause to explore in detail in your case. I then analyzed the data and, in this example, plotted it using a different approach. I obtained the same answers, which was gratifying. In the end, I confirmed that the data matched the reported results.

How Much?

I intended to write about 10 pages and ended up with 18. If you count it up, there is about six pages worth of figures and tables so I am pretty close to my goal of 10 pages of writing.

What to do Now?

The document above represents a second draft of a personal exploration. I need to go back and edit it now. Are there asides and commentary that do not serve the goals of the exploration? Are there places where I say the same thing twice? Can it be better organized for flow and clarity?

To properly copyedit the exploration, I must finish my second draft and let it sit for a few days. I can only proofread after I have forgotten what I had written. Otherwise, my mind fills in the correct things I meant to say and I overlook the errors.⁵⁷ Give yourself the time – start early.

⁵⁷ Did you see what I just did there?

A Note on References

I referenced 25 journal articles. Some were reviews, some were off-topic (the Prins reaction, for example), some merely provided further resources on a topic that was mentioned (e.g., the Diels-Alder reaction or the connection between entropy and symmetry), and a few were the core papers that were explored in detail. All publications are easily accessible to the reader via the DOI links provided. When you reference a paper, you must provide a link that will take the reader quickly to the resource. I will want to find the papers to confirm that they contained the information claimed. Choose quality papers, not just the latest paper that has words applying to your topic in its title. Remember, your textbook is a valuable reference. It makes me feel good when I believe you have been using it.

Have Fun

A personal exploration is not a formal thesis. Feel free to go down a rabbit hole. I wanted to present more about the Baird IR spectrometer and explain how it worked to generate the IR light (a glow plug?), how it separated the wavelengths of IR light (NaCl prisms?) and how the dual-beam setup enabled IR spectra of materials in solution. I could not find good information about the instrument within ten minutes of poking around the web, and so dropped that line of thought. It was just a fun side trip, after all. I referenced a review of the company's history that made the spectrometer, primarily so that I could pick up that thread later if I so chose. I spent way too much time on the bootstrap analysis. But the code is written now, and I can use it again someday. It was cool that former UPEI chemistry faculty members turned up in my searches about the Woodward-Hoffmann rules. Fun things pop up when you are looking around.

Just write and don't stop to consider whether you will keep it later. Cutting pages in the editing process is far easier than writing them. While the words are flowing, don't slow down. You can tidy things up later.

I hope that you find the exercise of exploring a literature topic fun and informative.